Module Code: CO822

Examination number: 199662

University of Kent

May/June 2021

Examination Answer Booklet

Before starting this examination, you **MUST enter your Examination Number** above **(double click)**; failure to provide this means your mark cannot be recorded and **you may be awarded a 0 (zero)** for this examination.

Re-save the answer booklet to include your exam number in the file name, using the following format: [**your exam number]** [**module code] AnswerBooklet** (e.g. *123456 XX123 AnswerBooklet*).

Answer one question per page. You may add additional pages to this document if necessary.

Question Number	1

Add your answer below:

a)

i) 10

ii) Length: 2*sqrt(11) and sqrt(3)

Normalised vectors: (1/sqrt(11), 1/sqrt(11), 3/sqrt(11)) and (1/sqrt(3), 1/sqrt(3), 1/sqrt(3))

5 / sqrt(33)

b)

Substitute $|n_i\rangle$ with $|\phi_i\rangle$:

$|\psi\rangle=\sum_{i,j} a_j c_j^i |\phi_j\rangle$

c)

$|\psi\rangle = \sum_i |e_i\rangle_1 \otimes |l_i\rangle_2$

$\rho 1 = tr (\sum_{i,k} \langle n_k|e_i\rangle_1 \otimes \langle n_k|l_i\rangle_2)$

$= \sum_{i,k} \langle n_k|e_i\rangle_1 \otimes \langle n_k|l_i\rangle_2$

$= \sum_{i,k} |e_i\rangle_1 |l_i\rangle_2$

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I attempted Schmidt decomposition to find the result because I assume the two states are entangled, as i is on both states.

d)

Question Number	2

Add your answer below:

a)

i) 1 with a probability of 100%
ii) 1 with a probability of 100%

b)

i) Probability = 1/2
ii) 0
iii) 0 or 1 (Probability = 1/2)
iv) Probability = 3/4

c)

i) Probability = 1/3
ii) Probability = 4/6
iii) This does improve security, as it decreases the probability of Eve correctly deciding what Alice sent. As having a computational (Z) and Hadamard (X) basis gives a 3/4 (75%) probability of Eve correctly deciding Alice sent a 0, with having an extra Y basis it decreases the probability to 4/6 (66%). Giving a 9% increase in security.